DELINEO · LOCAL SIMULATION STUDY

Delineo Weekly Update

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tl;dr: 56 completed engine runs · 80 invented people · 4 simulated days · 10 paired seeds

Engine 693e98c · October 2026

Workflow

Instead of using the API or website, we built a workaround.

1 Read the prior work

Chang et al., Nature 2021, the KDD policy tool, and the hosted Barnsdall notebook (runs 187–196) [past runs].

2 Found gaps

Denied visits have no household. Rankings are one metapopulation draw. Reopening only edits mobility. Hosted runs did not keep the schedule that was registered.

3 Ran the engine locally

56 new folders/runs on the delineo repository via an engine we built.

4 Measured the gaps

Same visit-hours removed two ways. Home infections counted separately. Ten seeds, same index cases.

Gaps

Two sources: the Nature mobility model, and the earlier Delineo website runs.

Source	What it established	What it does not resolve
Chang et al. 2021	A peak-occupancy cap prevents more infections than deleting the same visits evenly. A few venues dominate.	A denied visit leaves the network. It is not reassigned to a household. People are census-block totals, not individuals.
Chang et al. FAQ and KDD 2021	Thousands of mobility counterfactuals can be scanned for a metro area.	Curfews, coming back later, and which real cause produces the new mobility were not tested. SafeGraph misses children and some venues.
Hosted Barnsdall notebook	School and venue hotspots are a useful question. Repeats were requested because seeds differ.	Initial infections differed, so runs were not paired.

Solutions

Invented population. Identical schedules. The only difference is floor area.

80 people

20 households of four. One index case in each group, persons 0 and 40.

Two halls

40 people, same hours. Small hall entered as 50 m². Large hall 1,000 m².

Four days

Peaks at 08:00, 12:00, and 17:00 (40 people). Seven other daytime hours have 8.

Five policies

Baseline, 20% cap, matched uniform cut, close the small hall, mask from hour 24.

VISIT BURDEN

1,408

venue person-hours at baseline

640

under both the cap and the uniform cut. Same hours removed.

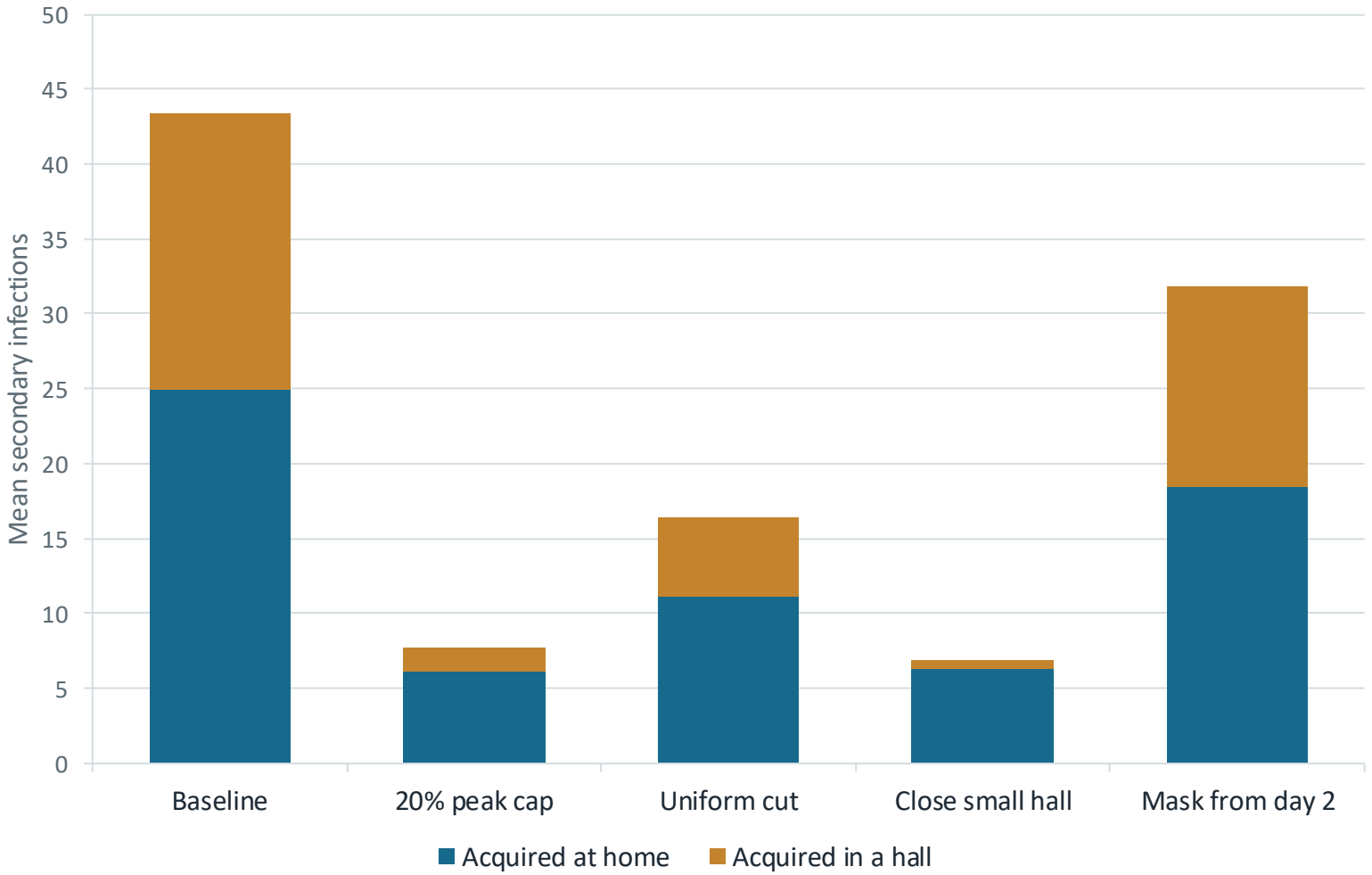
704

after closing only the small hall. The large hall stays open.

The engine clamps areas under 65 m² up to 65 m², so the small hall was simulated at 65, not 50.

2ndary infections

Mean over seeds 101–110. Index cases excluded. Home and hall counts add up in every run.



Notable Notes

43.4
baseline secondary infections. 25.0 of them at home.

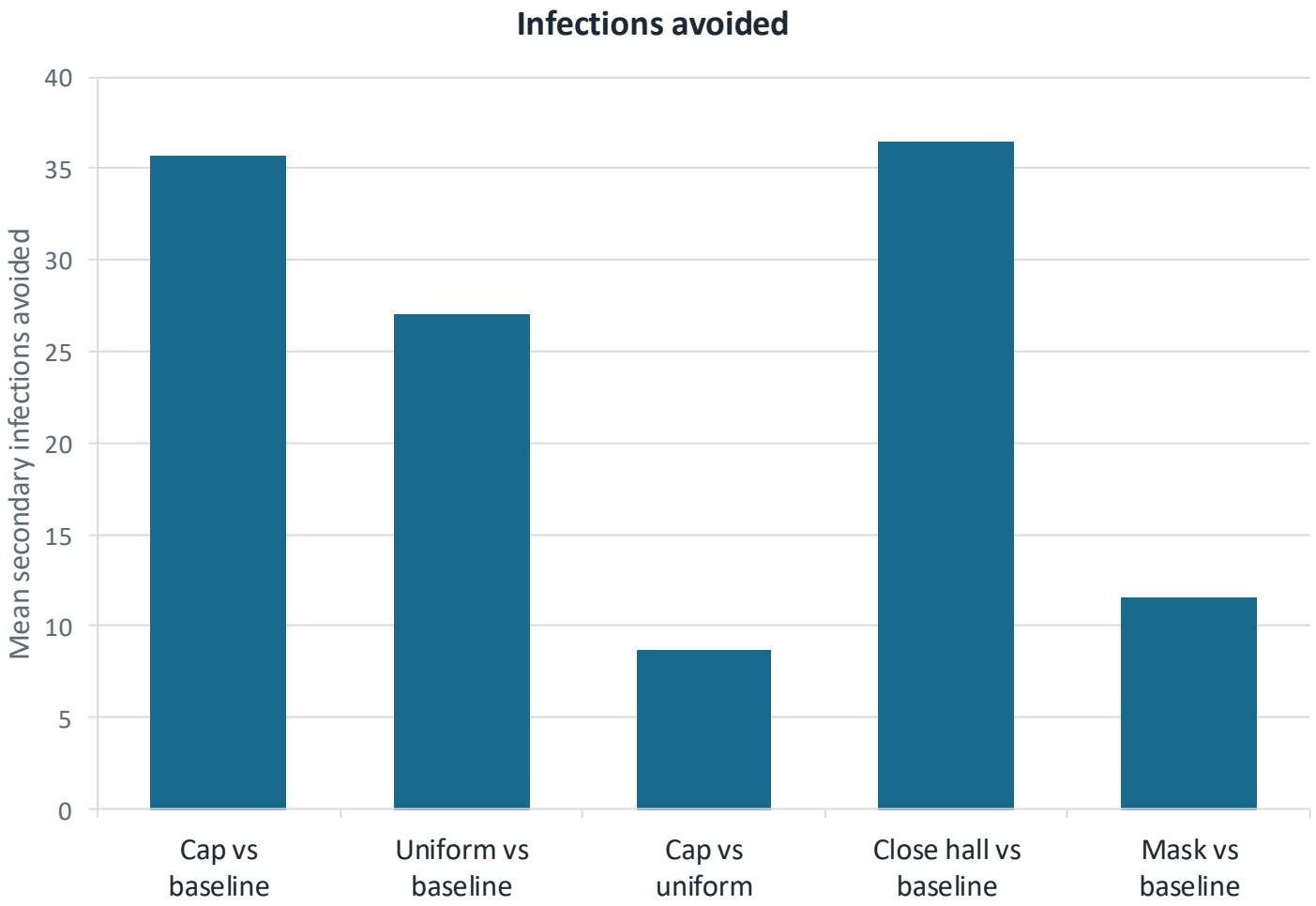
7.7
under the cap. Hall infections fall from 18.4 to 1.6.

16.4
under the uniform cut that removes the same hours.

6.9
after closing the small hall. 6.3 of those are at home.

The Chang comparison by pairing seeds

Positive numbers are infections avoided. Intervals are the 10-seed bootstrap, not parameter uncertainty.



95% INTERVAL · SEEDS STRICTLY LOWER

Cap vs baseline
32.9 – 38.8 10 of 10

Uniform vs baseline
23.1 – 32.0 10 of 10

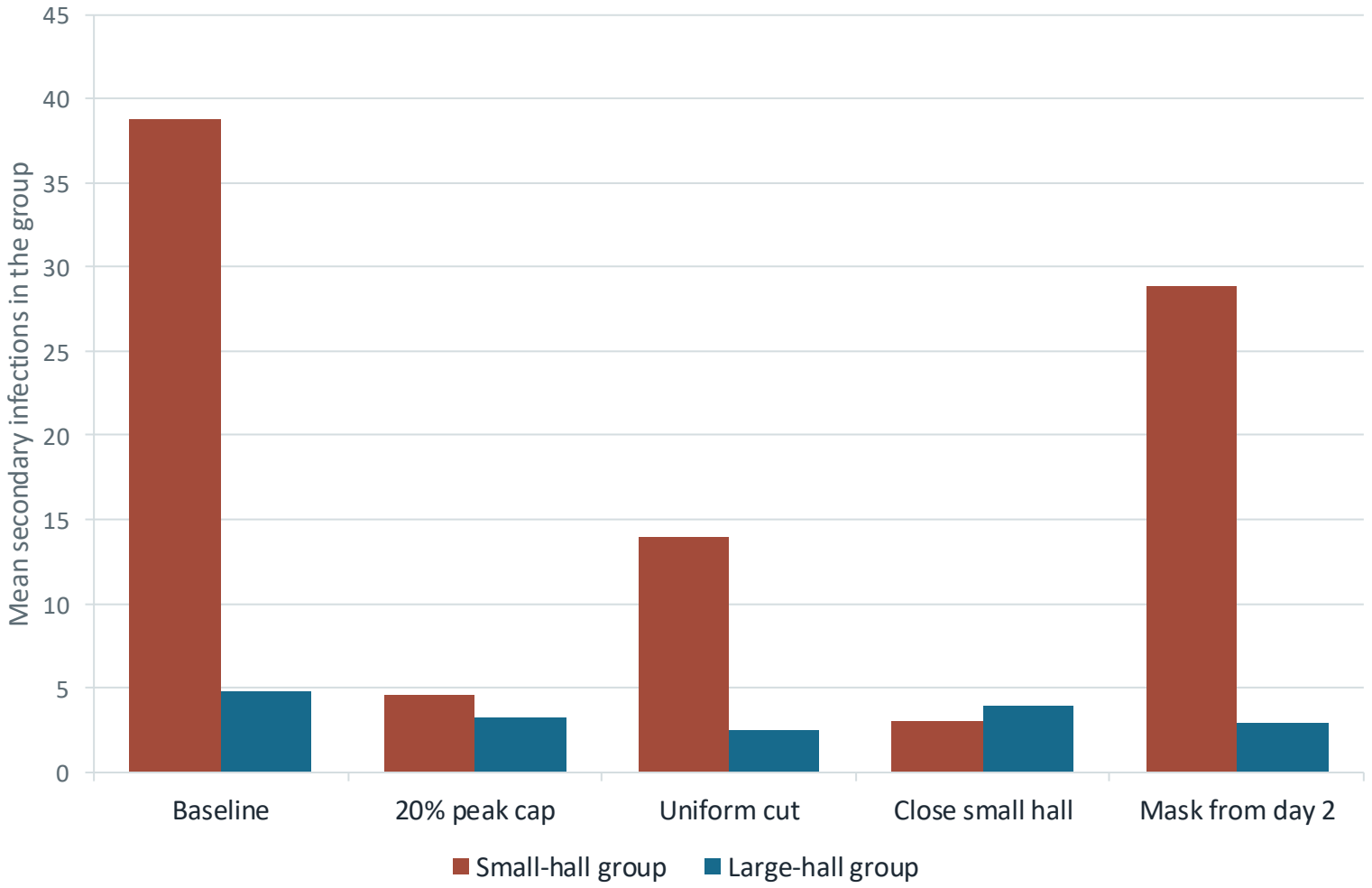
Cap vs uniform
5.6 – 11.6 9 of 10

Close small hall
35.3 – 38.1 10 of 10

Mask from hour 24
6.6 – 17.5 10 of 10

Group gap

This is the crowded-venue mechanism from Chang et al., with income and race left out.



CHANGES

38.7 vs 4.7

Baseline. Almost the whole small-hall group was infected. The large-hall group was not.

4.5 vs 3.2

Under the cap. The group gap mostly disappears. The extra benefit of the cap over uniform thinning is in the crowded group (9.4 infections, interval 6.3–12.3).

How each gap was addressed

The result column is the evaluation set only. Seed 100 was a discovery check and is excluded.

GAP	WHAT THE RUN DID	WHAT THE 10 SEEDS SHOWED
Denied visit has nowhere to go	Overflow and closures stay home. Home infections are a separate count.	Cap cut home infections by 18.9 and hall infections by 16.8. Closing the hall left 6.3 of 6.9 remaining infections at home.
One draw decides the ranking	Ten paired seeds, same index cases, mobility frozen.	Cap beat the matched uniform cut on 9 seeds and tied seed 102 at 3. It was never worse.
Equity mixes density with who can stay home	Both groups travel the same hours. Only the room size changes.	Baseline 38.7 vs 4.7. Cap 4.5 vs 3.2. No income or race field was used.
Reopening analysis only edits visits	Mask about half the population from hour 24, on the baseline trace.	Masking avoided 11.6 infections (6.6–17.5), less than either mobility policy, with a wider interval.
Website runs were not paired	Local runner. Hashed inputs. New folder per run. Network blocked.	50 evaluation runs plus a zero-seed control that stayed uninfected for all 96 hours.

Complications

The policy ranking is for this clock and this population.

Future

Disease clock is the 4-hour / 24-hour fallback. DMP required is the higher-fidelity model. A 3-day required-mode check produced no secondary infections, so a 4-day comparison would not have shown a policy effect.

Not calibrated to cases. Ventilation coefficient 9 is the engine default, not a fitted value.

People who lose a visit go home. They do not switch halls or return later.

Ten seeds describe Monte Carlo error. They do not estimate a city.

Collaborators

Result metadata says deterministic:0 even when an explicit seed was used.

Default DMP mode is auto. A missed match still succeeds, on a 24-hour illness.

Rooms under 65 m^2 are simulated as 65 m^2, and the result does not record the clamped area.

Households keep a fixed ventilation of 150 m^3/hr. Is this intentional?

Pipeline

1

prepare

Writes pap.json, three movement files, and five policy configs.

2

discover

Seed 100 only. Confirmed the small hall was the hotspot. Excluded from the means.

3

zero

Baseline with no index case. Stayed at zero infections.

4

evaluate

Seeds 101–110, five policies. Fifty new output folders.

5

analyze

Keeps a run only if its manifest says complete. Writes the tables behind these slides.

Writeup

docs/chang-gap-study.md

Tables

results/chang-gap/

Each run

runs/chang-gap/<arm>-<seed>/

About 0.6 seconds and 150 MiB per run. The runner validates inputs, blocks network calls, records the engine commit and file hashes, then calls SimulationRunner.

Rerun from the delineo folder: python scripts/chang_gap_study.py evaluate --seeds 101-110